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HALF YEARLY EXAMINATION - 2025

CLASS: XII **MATHEMATICS** Reg.No

Time : 3.00 Hours **MARKS : 90**

PART – I

Answer all the questions . **20 x 1 = 20**

1. If $\rho(A) = \rho([A|B])$, then the system AX = B of linear equations is
   1) consistent and has a unique solution 2) consistent
   3) consistent and has infinitely many solution 4) inconsistent

2. If $A=\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then λ is
   1) 17 2) 14 3) 19 4) 21

3. If $\omega \neq 1$ is a cubic root of unity and $(1+\omega)^7 = A + B\omega$, then (A, B) equals
   1) (1, 0) 2) (−1, 1) 3) (0, 1) 4) (1, 1)

4. If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is
   1) 1 2) 2 3) 3 4) 4

5. If α, β and γ are the roots of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is
   1) $-\frac{q}{r}$ 2) $-\frac{p}{r}$ 3) $\frac{q}{r}$ 4) $-\frac{q}{p}$

6. The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is
   1) [1, 2] 2) [−1, 1] 3) [0, 1] 4) [−1, 0]

7. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is
   (1) 1 (2) 3 (3) $\sqrt{10}$ (4) $\sqrt{11}$

8. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
   (1) $2ab$ (2) ab (3) $\sqrt{ab}$ (4) $\frac{a}{b}$

9. The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is
   (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{2}$

10. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j}$, $\vec{c} = \hat{i}$ and $(\vec{a} \times \vec{b}) \times \vec{c} = \lambda\vec{a} + \mu\vec{b}$, then the value of $\lambda + \mu$ is
    (1) 0 (2) 1 (3) 6 (4) 3

11. The function $\sin^4 x + \cos^4 x$ is increasing in the interval
    (1) $\left[\frac{5\pi}{8}, \frac{3\pi}{4}\right]$ (2) $\left[\frac{\pi}{2}, \frac{5\pi}{8}\right]$ (3) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (4) $\left[0, \frac{\pi}{4}\right]$

12. The volume of a sphere is increasing in volume at the rate of 3π cm^3 / sec. The rate of change of its radius when radius is $\frac{1}{2}$ cm
    (1) 3 cm/s (2) 2 cm/s (3) 1 cm/s (4) $\frac{1}{2}$ cm/s

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13. Linear approximation for $g(x) = \cos x$ at $x = \frac{\pi}{2}$ is

(1) $x + \frac{\pi}{2}$ (2) $-x + \frac{\pi}{2}$ (3) $x - \frac{\pi}{2}$ (4) $-x - \frac{\pi}{2}$

14. The value of $\int_0^{\pi} \sin^4 x\,dx$ is

1) $\frac{3\pi}{10}$ 2) $\frac{3\pi}{8}$ 3) $\frac{3\pi}{4}$ 4) $\frac{3\pi}{2}$

15. The order and degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{1/3} + x^{1/4} = 0$ are respectively

(1) 2, 3 (2) 3, 3 (3) 2, 6 (4) 2, 4

16. Integrating factor of the differential equation $\frac{dy}{dx} = \frac{x+y+1}{x+1}$ is

(1) $\frac{1}{x+1}$ (2) $x+1$

(3) $\frac{1}{\sqrt{x+1}}$ (4) $\sqrt{x+1}$

17. The probability function of a random variable is defined as:

x	-2	-1	0	1	2
f(x)	k	2k	3k	4k	5k

Then E(X) is equal to

(1) $\frac{1}{15}$ (2) $\frac{1}{10}$ (3) $\frac{1}{3}$ (4) $\frac{2}{3}$

18. Which one of the following is incorrect? For any two propositions p and q, we have

(1) $\neg(p \vee q) \equiv \neg p \wedge \neg q$ (2) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

(3) $\neg(p \vee q) \equiv \neg p \vee \neg q$ (4) $\neg(\neg p) \equiv p$

19. If the function $f(x) = \frac{1}{100}$ for $a < x < b$, represents a probability density function of a continuous random variable X, then which of the following can be the value of a and b?

(1) 0 and 120 (2) 100 and 200 (3) 1 and 100 (4) 160 and 240

20. In the set R of real numbers '*' is defined as follows. Which one of the following is not a binary operation on R ?

(1) $a * b = a^2 - b^2$ (2) $a * b = a^2 + b^2$ (3) $a * b = \sqrt{ab}$ (4) $a * b = \sqrt{a^2 + b^2}$

PART – II

Answer any seven questions. Question No.30 is compulsory. **7 x 2 = 14**

21. Find the rank of the matrix by minor method: $\begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$

22. Find the monic polynomial equation of minimum degree with real coefficients having $2 - \sqrt{3}\,i$ as a root.

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23. Simplify $\sin^{-1}(\sin 10)$

24. For any vector $\vec{a}$, prove that $\vec{i}\times(\vec{a}\times\vec{i})+\vec{j}\times(\vec{a}\times\vec{j})+\vec{k}\times(\vec{a}\times\vec{k})=2\vec{a}$

25. Evaluate the limit $\lim\limits_{x\to 0}\left(\dfrac{\sin x}{x^2}\right)$

26. Evaluate the following $\int\limits_0^{\infty} x^5 e^{-3x}\,dx$

27. Find the differential equation of the family of parabolas $y^2=4ax$, where a is an arbitrary constant.

28. If X is the random variable with distribution function $F(x)$ given by,

$$F(x)=\begin{cases}0 & -\infty<x<0\\ \frac{1}{2}(x^2+x) & 0\le x<1\\ 1 & 1\le x<\infty\end{cases}$$ then find the probability density function $f(x)$

29. Let $*$ be defined on R by $(a*b)=a+b+ab-7$. Is $*$ binary on R ? If so, find $3*\left(\frac{-7}{15}\right)$.

30. Show that $|3z+5-2i|=4$ represents a circle, and find its centre and radius.

PART – III

Answer any Seven questions. Question No.40 is compulsory **7 x 3 = 21**

31. If $A=\begin{bmatrix}3 & 2\\ 7 & 5\end{bmatrix}$ and $B=\begin{bmatrix}-1 & -3\\ 5 & 2\end{bmatrix}$, verify that $(AB)^{-1}=B^{-1}A^{-1}$.

32. Find the values of the real numbers x and y, if the complex numbers $(3-i)x-(2-i)y+2i+5$ and $2x+(-1+2i)y+3+2i$ are equal.

33. Find the value of $\tan^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$

34. Find the equation of the ellipse with foci $(\pm 2, 0)$, vertices $(\pm 3, 0)$.

35. Find the shortest distance between the two given straight lines $\vec{r}=\left(2\hat{i}+3\hat{j}+4\hat{k}\right)+t\left(-2\hat{i}+\hat{j}-2\hat{k}\right)$ and $\dfrac{x-3}{2}=\dfrac{y}{-1}=\dfrac{z+2}{2}$

36. Evaluate : $\lim\limits_{x\to 0^+}\left(\dfrac{1}{x}-\dfrac{1}{e^x-1}\right)$.

37. Show that the percentage error in the nth root of a number is approximately $\frac{1}{n}$ times the percentage error in the number.

38. Evaluate the following integrals using properties of integration : $\int\limits_{-5}^{5} x\cos\left(\dfrac{e^x-1}{e^x+1}\right)dx$

39. Solve: $\dfrac{dy}{dx}=e^{x+y}+x^3e^y$

40. The probability that Mr.Q hits a target at any trial is $\frac{3}{4}$. Suppose he tries at the target 15 times. Find the probability that he hits the target (i) exactly 5 times (ii) at least one time.

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PART – IV

Answer all the questions. 7 x 5 = 35

41. **(a)** Solve the following systems of linear equations by Cramer's rule:

$\frac{3}{x}-\frac{4}{y}-\frac{2}{z}-1=0,\ \frac{1}{x}+\frac{2}{y}+\frac{1}{z}-2=0,\ \frac{2}{x}-\frac{5}{y}-\frac{4}{z}+1=0.$ **(or)**

(b) If the curves $ax^2 + by^2 = 1$ and $cx^2 + dy^2 = 1$ intersect each other orthogonally then, show that $\frac{1}{a}-\frac{1}{b}=\frac{1}{c}-\frac{1}{d}$.

42. **(a)** $2\cos\alpha = x+\frac{1}{x}$ and $2\cos\beta = y+\frac{1}{y}$ show that

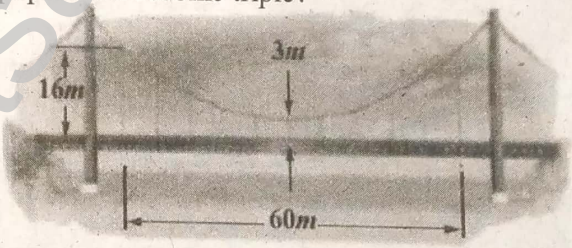
(i) $\frac{x^m}{y^n}-\frac{y^n}{x^m}=2i\sin(m\alpha-n\beta)$ (ii) $x^m y^n+\frac{1}{x^m y^n}=2\cos(m\alpha+n\beta)$ **(or)**

(b) If $v(x,y)=\log\left(\frac{x^2+y^2}{x+y}\right)$, prove that $x\frac{\partial v}{\partial x}+y\frac{\partial v}{\partial y}=1$.

43. **(a)** If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$ and $0<x, y, z<1$, show that $x^2+y^2+z^2+2xyz=1$. **(or)**

(b) The growth of a population is proportional to the number present. If the population of a colony doubles in 50 years, in how many years will the population become triple?

44. **(a)** Parabolic cable of a 60m portion of the roadbed of a suspension bridge are positioned as shown below. Vertical Cables are to be spaced every 6m along this portion of the roadbed. Calculate the lengths of first two of these vertical cables from the vertex. **(or)**



(b) A six sided die is marked '1' on one face, '3' on two of its faces, and '5' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find

(i) the probability mass function (ii) the cumulative distribution function

(iii) $P(4 \le X < 10)$ (iv) $P(X \ge 6)$

45. **(a)** We have a 24 square unit piece of thin material and want to make an open box by cutting small squares from the corners of our material and folding the sides up. The question is, which cut produces the box of maximum volume?. **(or)**

(b) Find the area of the region bounded between the parabola $x^2 = y$ and the curve $y = -x$.

46. **(a)** Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Also find the coordinates of the point of contact. **(or)**

(b) Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r}=\left(\hat{i}-\hat{j}+3\hat{k}\right)+t\left(2\hat{i}-\hat{j}+4\hat{k}\right)$ and perpendicular to plane $\vec{r}.\left(\hat{i}+2\hat{j}+\hat{k}\right)=8$.

47. **(a)** Prove $p\rightarrow(q\rightarrow r)\equiv(p\wedge q)\rightarrow r$ using truth table. **(or)**

(b) Prove by vector method that $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$.

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